

034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

FULL WRITTEN EXAMINATION - A.Y. 2023/2024

September 2, 2024

PART 2 - Design of a Controller

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Control of the Temperature in a Chemical Reactor

First Approach: a Dual-Loop Scheme

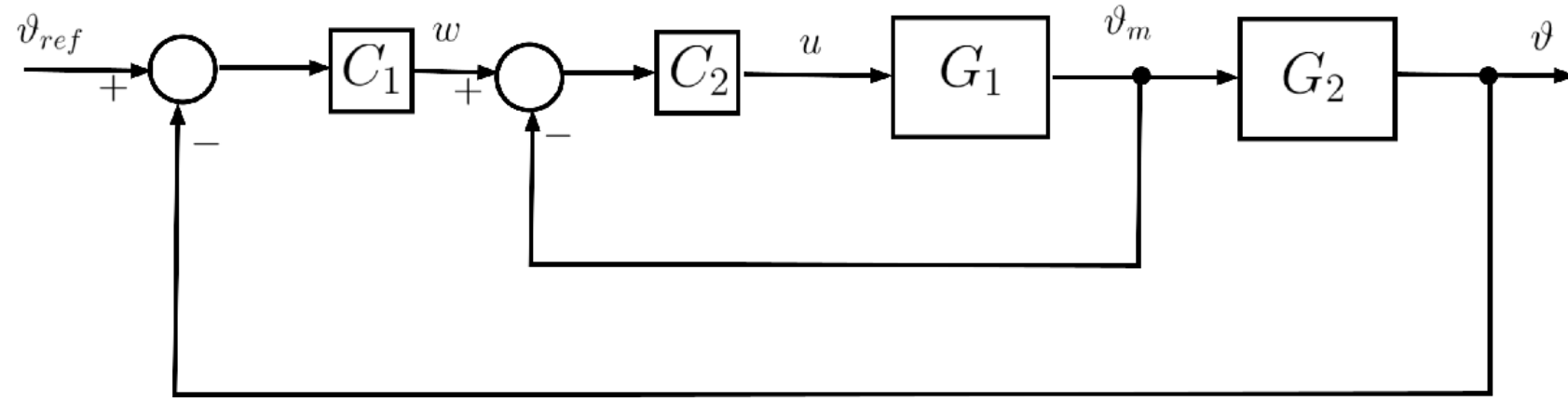


Figure 1: dual-loop control scheme for the temperature control in a chemical reactor.

To control the temperature $\vartheta(t)$ in a chemical reactor, the control system in [Figure 1](#) is used, where $\vartheta_{\text{ref}}(t)$ is the reference temperature (the desired temperature). The temperatures $\vartheta(t)$ and $\vartheta_m(t)$ in the reactor and in the cooler, respectively, are measurable and can be used to control the valve $u(t)$.

- The **inner loop** regulates the cooler's temperature, $\vartheta_m(t)$, using $w(t)$ as the reference target.
- The inner loop control variable is the cooling valve position $u(t)$.
- The **outer loop** aims to regulate the temperature $\vartheta(t)$ of the fluid in the reactor, with $\vartheta_{\text{ref}}(t)$ as the reference temperature.
- The control variable $w(t)$ is acting as the reference for the inner control loop.

It is here assumed that both $C_1(s)$ and $C_2(s)$ are proportional controllers:

$$C_1(s) = k_1, \quad C_2(s) = k_2, \quad k_1, k_2 \in \mathbb{R}$$

Finally

$$G_1(s) = \frac{1}{10s + 1}, \quad G_2(s) = \frac{1}{(30s + 1)(3s + 1)}$$

Assignment

A1: Let $k_2 = +19$ and carry out the following steps:

1. Compute the transfer function from $w(t)$ to $\vartheta_m(t)$. Let identify such a transfer function as $F_2(s)$.
2. Draw the Bode plot (both the asymptotic and the effective) of the frequency response of $F_2(s)$.
3. Compare the Bode plots of $F_2(j\omega)$ with those of $G_2(j\omega)$. Any remark?
4. Choose the gain k_1 so that the gain margin (of the outer open-loop transfer function) results $K_{m1} = 12$ dB.
5. Determine the crossover frequency ω_{c1} , the phase margin φ_{m1} , and the steady-state error, assuming that the temperature reference is a stepwise function: $\vartheta_{\text{ref}}(t) = \mathbf{1}(t)$

As usual, $\mathbf{1}(t)$ denotes the unit step function: $\mathbf{1}(\tau) = \begin{cases} 0 & \longleftrightarrow \tau < 0 \\ 1 & \longleftrightarrow \tau \geq 0 \end{cases}$

HINT: in order to fulfil assignment A1.4 follow this approach

- Firstly determine the angular frequency ω_π , such that $\arg [F_2(j\omega_\pi) \cdot G_2(j\omega_\pi)] = -180^\circ$. It does not depend on the gain k_1 , doesn't it?
- Evaluate the amplitude of $L_1(j\omega_\pi)$, with $L_1(s) = F_2(s) \cdot G_2(s)$, i.e. the outer open-loop transfer function. That amplitude has to depend on the gain k_1 .
- Determine the proper value for the gain k_1 , in order to guarantee the gain margin $K_{m1} = 12$ dB.

Second Approach: a Single-Loop Scheme

Consider now the following control system scheme, with a simple feedback loop.:

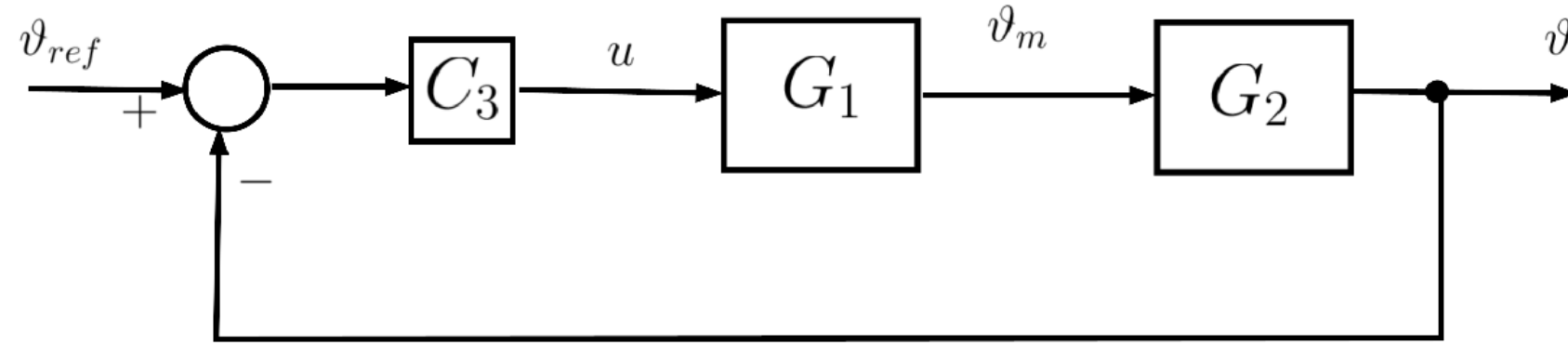


Figure 2: a simple feedback loop for the temperature control in the chemical reactor.

The controller $C_3(s)$ is a simple proportional controller

$$C_3(s) = k_3, \quad k_3 \in \mathbb{R}$$

and

$$G_1(s) = \frac{1}{10s + 1}, \quad G_2(s) = \frac{1}{(30s + 1)(3s + 1)}$$

As in the previous scenario, $\vartheta_{\text{ref}}(t)$ is the reference temperature (the desired temperature). The variables $\vartheta(t)$ and $\vartheta_m(t)$ are the temperatures in the reactor and in the cooler, respectively. The variable $u(t)$ is the position of the cooling valve.

Assignment

A2: Carry out the following steps:

1. Compute the open-loop transfer function and identify such a transfer function as $\hat{L}(s)$.
2. How is the Bode plot affected? Draw the Bode plot (both the asymptotic and the effective) of the frequency response of $\hat{L}(s)$.
3. Compare the Bode plots of $\hat{L}(j\omega)$ with those obtained, in the previous scenario, for the outer loop open-loop transfer function, i.e. $L_1(j\omega) = F_2(j\omega) \cdot G_2(j\omega)$. Any remark?
4. Choose the gain k_3 so that the gain margin results $\hat{K}_m = 12$ dB.
5. Determine the crossover frequency $\hat{\omega}_c$, the phase margin $\hat{\varphi}_m$, and the steady-state error, assuming that the temperature reference is a stepwise function: $\vartheta_{\text{ref}}(t) = \mathbf{1}(t)$.

6. **Compare** with the results in the [scenario A1](#). **Conclusions?**

As usual, $\mathbf{1}(\tau)$ denotes the unit step function: $\mathbf{1}(\tau) = \begin{cases} 0 & \longleftrightarrow \tau < 0 \\ 1 & \longleftrightarrow \tau \geq 0 \end{cases}$

Foreword to the Provided Solution

The live script is very long. In fact, we propose two different approaches to fulfil the request for a few parts of the solution. Moreover, we insert some more plots than the ones required to better clarify the solution approach.

A Practical Advise

Due to the length of the script, the suggested practical approach is to analyse and execute the code section by section, using the commands "Run Section" or "Run and Advance".

Solution

First Approach: a Dual-Loop Scheme

```
% --- Live Script Initialisation ---
clear
close all
clc

% adding folders (and subfolders) to search path
addpath(genpath('BodeDiagram/'))

Bcolors = [0 0.4470 0.7410; ...
           0.9290 0.6940 0.1250; ...
           0.4940 0.1840 0.5560; ...
           0.4660 0.6740 0.1880; ...
           0.3010 0.7450 0.9330; ...
           0.8500 0.3250 0.0980; ...
           0.6350 0.0780 0.1840]; % some different colors
                                % for the actual diagrams
```

A1: Design of the Outer Loop Controller

Let's define the transfer function of the processes $G_1(s)$, $G_2(s)$ and configure the MATLAB Search Path, by adding the folder containing the M-code able to compute and plot the asymptotic approximation of the Bode diagrams of the frequency response of a given transfer function. Finally, let's assign the inner-loop controller, as required in A1.1.

```
% let's define the transfer function builder element
s=tf('s');

% --- the process transfer functions G1s & G2s ---
G1s = 1/(10*s+1);           % the process G1(s)
G2s = 1/((30*s+1)*(3*s+1)); % the process G2(s)
```

A1.1 - The Inner Loop Controller: A Simple Proportional Gain

```
% --- the inner loop controller C2s ---
K2 = +19;
C2s = tf(K2);
```

The Inner-Loop Complementary Sensitivity Function

As requested in A1.1, let's compute the closed-loop transfer function $F_2(s)$ of the inner loop

```
% let's compute the open-loop TF of the inner loop L2s
L2s = C2s*G1s;    % the open-loop TF
```

```
F2s = feedback(L2s, 1, -1)
```

```
F2s =
```

```
      19
-----
10 s + 20
```

```
Continuous-time transfer function.
Model Properties
```

A1.3 - Comparing the Bode Plots

As requested in A1.1, let's compare the Bode plots of $F_2(j\omega)$ and $G_2(j\omega)$:

```
% --- A1.1: F2(j omega) vs G2(j omega) ---
hf1 = figure('Units','centimeters', 'Position',[0.1,0.1, 26, 26]);

omVALS = logspace(-2, 2, 1e5); % ang. freqs in [1/100; 100]
[hax1, hax2] = drawBodediagrams(F2s, omVALS, Bcolors(3, :), 2.5, '-', ...
                                Bcolors(6, :), 1.5, '-', hf1); % assignment A1.2

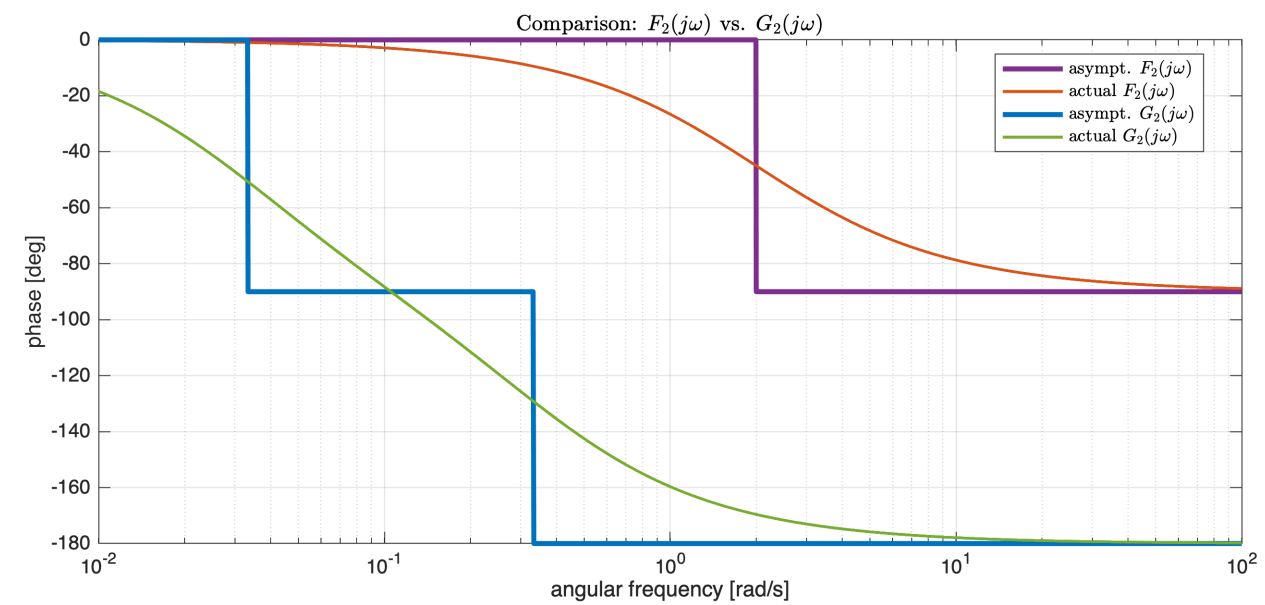
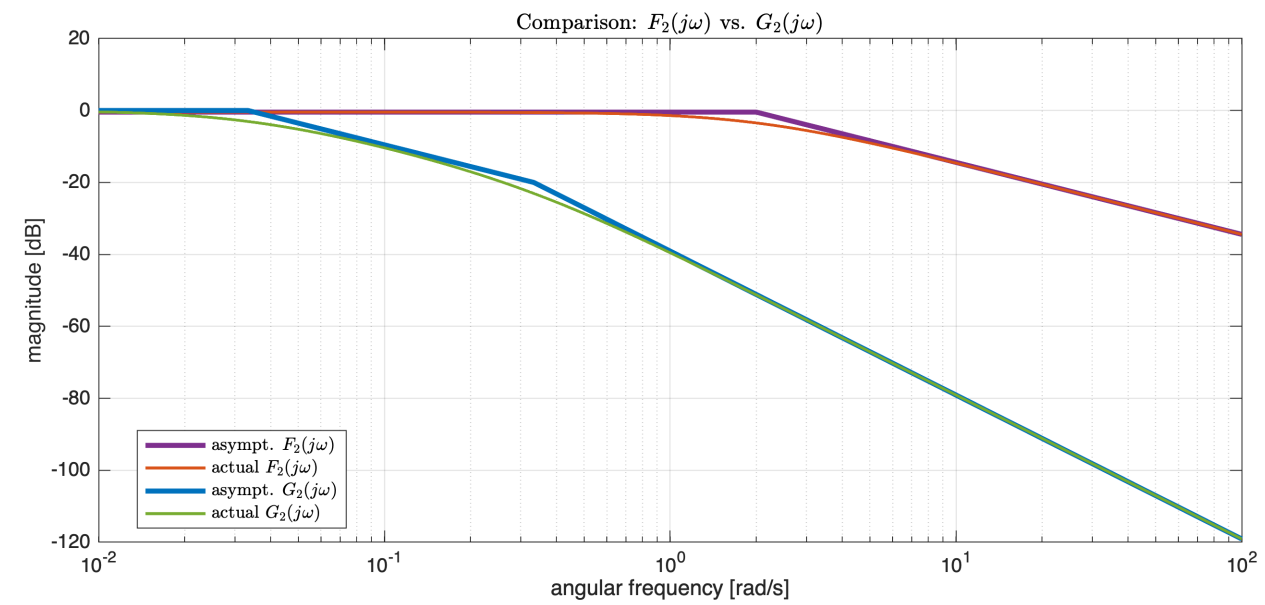
drawBodediagrams(G2s, omVALS, Bcolors(1, :), 2.5, '-', ...
                 Bcolors(4, :), 1.5, '-', hf1);
```

```

title(hax1, 'Comparison:  $F_2(j\omega)$  vs.  $G_2(j\omega)$ ', 'Interpreter', 'latex');
title(hax2, 'Comparison:  $F_2(j\omega)$  vs.  $G_2(j\omega)$ ', 'Interpreter', 'latex');

legend(hax1, 'asympt.  $F_2(j\omega)$ ', 'actual  $F_2(j\omega)$ ', ...
          'asympt.  $G_2(j\omega)$ ', 'actual  $G_2(j\omega)$ ', 'Location', 'best', ...
          'Interpreter', 'latex');
legend(hax2, 'asympt.  $F_2(j\omega)$ ', 'actual  $F_2(j\omega)$ ', ...
          'asympt.  $G_2(j\omega)$ ', 'actual  $G_2(j\omega)$ ', 'Location', 'best', ...
          'Interpreter', 'latex');

```



Remark: you can note that

- The crossover frequency ω_c for $F_2(s)$ is larger than the crossover frequency for $G_2(s)$: this **feature** is **effective** in **keeping** the **design** of the outer loop controller **easy**. You can approximate the transfer function $F_2(s)$ with an algebraic term (the static gain) when considering the frequencies from zero till the crossover frequency of $G_2(s)$.
- Moreover, considering the series connection of $F_2(s)$ and $G_2(s)$, i.e. $L_1(s) = F_2(s) \cdot G_2(s)$, you note that the crossover frequency ω_{cL_1} and the phase margin φ_{mL_1} are loosely affected by $F_2(s)$, with respect to the crossover frequency and the phase margin obtained whiit $G_2(s)$ only

fare grafici

A1.4 - Tuning the Outer-Loop Controller

To properly tune the gain k_1 (i.e. the outer loop controller $C_1(s)$), let's determine the outer-loop open-loop transfer function $L_1(s)$, i.e.

$$L_1(s) = C_1(s) \cdot F_2(s) \cdot G_2(s) \implies L_1(s) = k_1 \cdot \frac{19}{10s + 20} \cdot \frac{1}{(30s + 1)(3s + 1)} = \frac{19k_1}{(10s + 20)(30s + 1)(3s + 1)} = \frac{\frac{19}{20}k_1}{(0.5s + 1)(30s + 1)(3s + 1)}$$

As preliminar check, let's draw the Bode plots for $L_1(j\omega)$:

```
K1 = +1;
C1s = tf(K1);

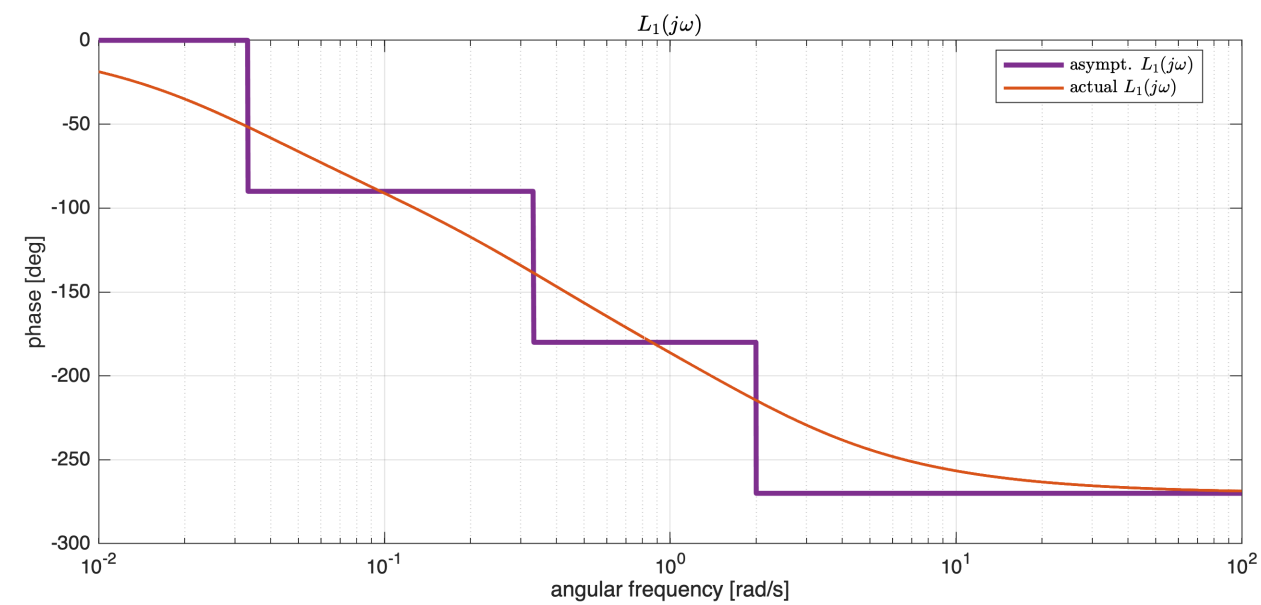
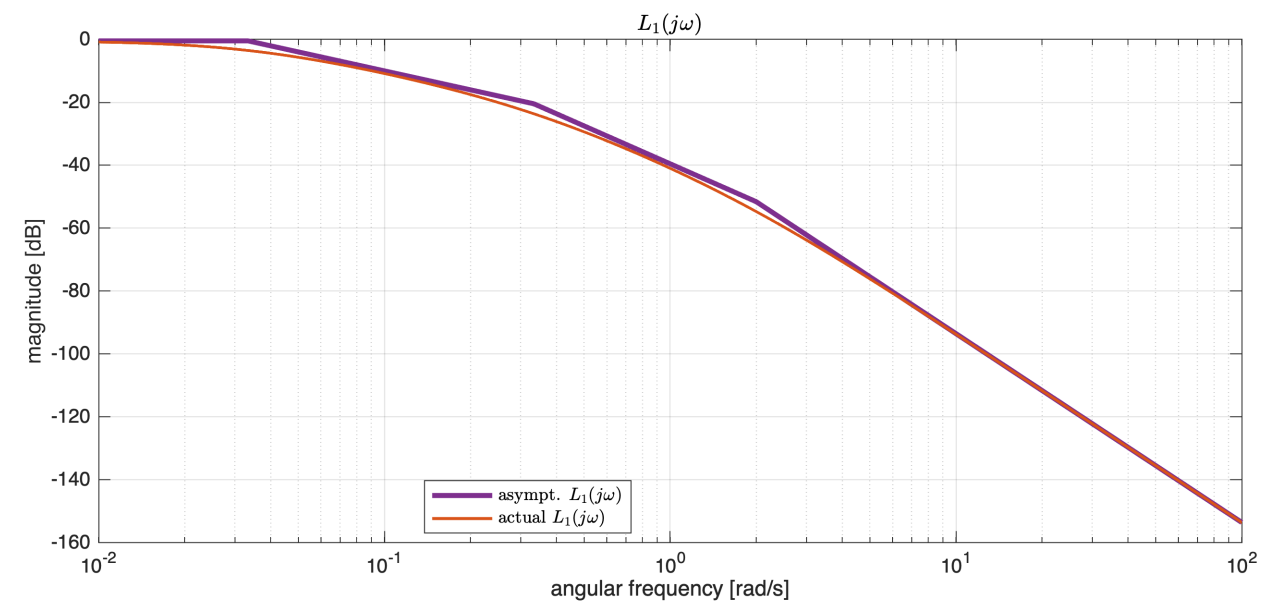
L1s = C1s*F2s*G2s;

hf2 = figure('Units','centimeters', 'Position',[0.1,0.1, 26, 26]);

omVALS = logspace(-2, 2, 1e5); % ang. freqs in [1/100; 100]
[hax3, hax4] = drawBodediagrams(L1s, omVALS, Bcolors(3, :), 2.5, '-', ...
                                Bcolors(6, :), 1.5, '-', hf2); % assignment A1.2

title(hax3, '$L_1(j \omega)$', 'Interpreter', 'latex');
title(hax4, '$L_1(j \omega)$', 'Interpreter', 'latex');

legend(hax3, 'asympt. $L_1(j \omega)$', 'actual $L_1(j \omega)$', ...
        'Location', 'best', ...
        'Interpreter', 'latex');
legend(hax4, 'asympt. $L_1(j \omega)$', 'actual $L_1(j \omega)$', ...
        'Location', 'best', ...
        'Interpreter', 'latex');
```



Analysing the Bode plots, we can assert that the angular frequency ω_π exists for sure, and $\omega_\pi < 1$ rad/s. How to compute ω_π ?

First approach: exploiting the Matlab command `margin( )`

```
[~, ~, omega_pi_approx, ~] = margin(L1s)
```

```
omega_pi_approx =  
0.8628
```

```
rad2deg(angle(freqresp(L1s, omega_pi_approx)))
```

```
ans =  
179.9994
```


Second approach: applying some mathematical manipulations.

As suggested in A1.4, let's calculate the angular frequency ω_π as

$$\omega_\pi : \arg L_1(j\omega_\pi) = -180^\circ \iff \arg\left(\frac{19}{20}k_1\right) - \arg[(0.5 j\omega_\pi + 1)(30 j\omega_\pi + 1)(3 j\omega_\pi + 1)] = -180^\circ$$

Assuming $k_1 > 0$, we obtain

$$\omega_\pi : \arg L_1(j\omega_\pi) = -180^\circ \iff -\arg[(0.5 j\omega_\pi + 1)(30 j\omega_\pi + 1)(3 j\omega_\pi + 1)] = -180^\circ$$

Thus

$$\omega_\pi : \arg L_1(j\omega_\pi) = -180^\circ \iff \arg[(0.5 j\omega_\pi + 1)] + \arg[(30 j\omega_\pi + 1)] + \arg[(3 j\omega_\pi + 1)] = 180^\circ$$

$$\arg[(30 j\omega_\pi + 1)] + \arg[(3 j\omega_\pi + 1)] = 180^\circ - \arg[(0.5 j\omega_\pi + 1)]$$

Since $\omega_\pi > 0$

$$\arctan\left(\frac{30\omega_\pi}{1}\right) + \arctan\left(\frac{3\omega_\pi}{1}\right) = 180^\circ - \arctan\left(\frac{0.5\omega_\pi}{1}\right) \iff \arctan(30\omega_\pi) + \arctan(3\omega_\pi) = 180^\circ - \arctan(0.5\omega_\pi)$$

Given the trigonometric identities

$$\tan(\alpha + \beta) = \frac{\tan(\alpha) + \tan(\beta)}{1 - \tan(\alpha)\tan(\beta)} \quad \alpha + \beta \neq 90^\circ + n180^\circ, n \in \mathbb{Z}$$

and

$$\tan(180^\circ - \delta) = -\tan \delta$$

we obtain

$$\arctan(30\omega_\pi) + \arctan(3\omega_\pi) = 180^\circ - \arctan(0.5\omega_\pi) \iff \frac{33\omega_\pi}{1 - 90\omega_\pi^2} = -0.5\omega_\pi$$

Assume

$$\omega_\pi \neq 0, \quad 1 - 90\omega_\pi^2 \neq 0, \quad \omega_\pi > 0$$

So

$$\frac{33\omega_\pi}{1 - 90\omega_\pi^2} = -0.5\omega_\pi \iff \frac{33}{1 - 90\omega_\pi^2} = -0.5 \iff \frac{33}{0.5} = -(1 - 90\omega_\pi^2) \iff 66 = 90\omega_\pi^2 - 1 \iff 90\omega_\pi^2 = 67 \iff \omega_\pi^2 = \frac{67}{90}$$

Finally

$$\omega_\pi = +\sqrt{\frac{67}{90}}$$

syms **omega**

```
assume(omega >=0)

OMeq = omega^2 == (67/90);
omega_pi = solve(OMeq, omega)
```

```
omega_pi =

$$\frac{\sqrt{670}}{30}$$

```

```
omega_pi_val = double(omega_pi)
```

```
omega_pi_val =
0.8628
```

```
rad2deg(angle(freqresp(L1s,omega_pi_val )))
```

```
ans =
-180
```

Now, let's evaluate the **magnitude** of $L_1(j\omega_\pi)$:

Assuming $k_1 = 1$ we can calculate the magnitude of the frequency response $|L_1(j\omega_\pi)|$ as follows

```
magL1 = abs(freqresp(L1s,omega_pi_val ))
```

```
magL1 =
0.0121
```

To fulfil A1.4, we have to increase $|L(j\omega_\pi)|$:

```
gainM_target_dB = +12; % dB
gainM_target = 10^(gainM_target_dB/20)
```

```
gainM_target =
3.9811
```

In fact (refer to Part 9 of the course material, slide 24)

$$K_m = \frac{1}{|L(j\omega_\pi)|}$$

Thus, to obtain $\bar{K}_m \text{ dB} = +12 \text{ dB}$, the magnitude $|\bar{L}_1(j\omega_\pi)|$ should have the value

$$|\bar{L}_1(j\omega_\pi)| = \frac{1}{\bar{K}_m} \approx \frac{1}{4} \implies \bar{k}_1 = \frac{|\bar{L}_1(j\omega_\pi)|}{|L_1(j\omega_\pi)|}$$

```
K1_A1_4 = (1/4)/magL1
```

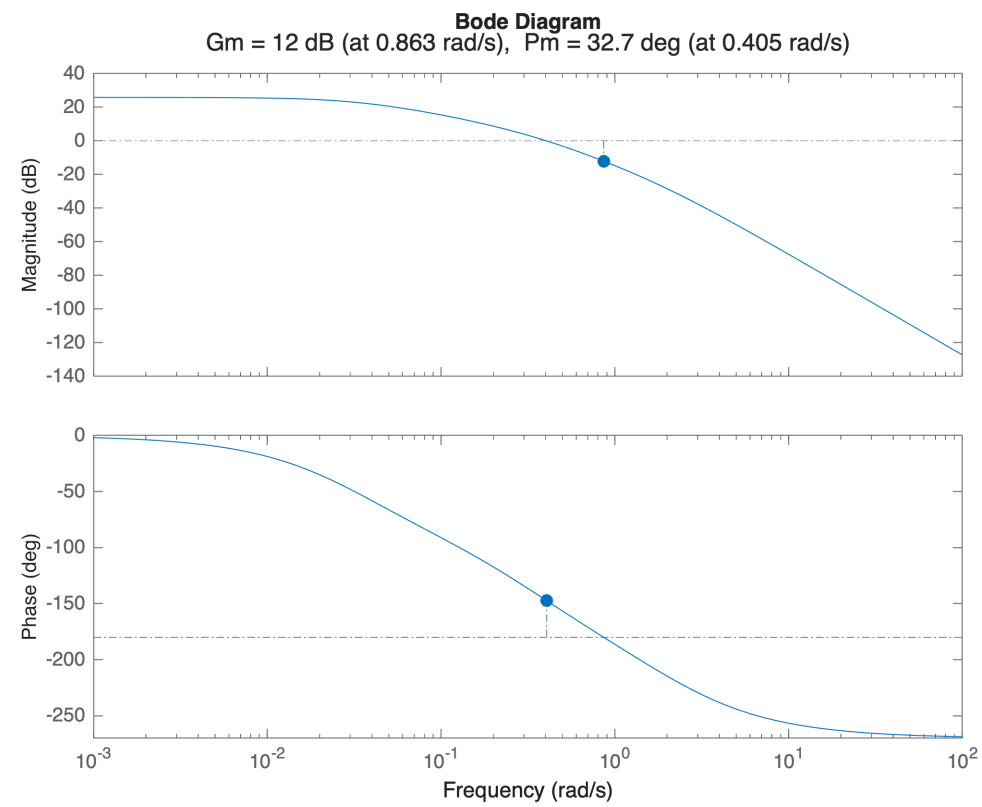
```
K1_A1_4 =
20.6009
```

Let's check the effective gain margin, when using this value for the gain k_1 :

```
C1s_A1_4 = tf(K1_A1_4);
```

```
L1s_A1_4 = C1s_A1_4*F2s*G2s;
```

```
figure; margin(L1s_A1_4)
```



A1.5: Crossover Frequency and Phase Margin

Simply

```
[gainM_2loop, phaseM_2loop, OM_GM_2loop, OM_PM_2loop]= margin(L1s_A1_4)
```

```
gainM_2loop =  
4.0001  
phaseM_2loop =  
32.7168  
OM_GM_2loop =  
0.8628  
OM_PM_2loop =  
0.4050
```

First Approach: a Recap

```
% --- the inner loop controller C2s ---  
K2 = +19;  
C2s = tf(K2);  
% let's compute the open-loop TF of the inner loop L2s  
L2s = C2s*G1s;    % the open-loop TF  
  
F2s = feedback(L2s, 1, -1);  
  
% --- the outer loop controller C1s ---  
C1s_A1_4 = tf(K1_A1_4);
```

```

L1s_A1_4 = C1s_A1_4*F2s*G2s;

% ----- final Bode plots -----
hf3 = figure('Units','centimeters', 'Position',[0.1,0.1, 26, 26]);

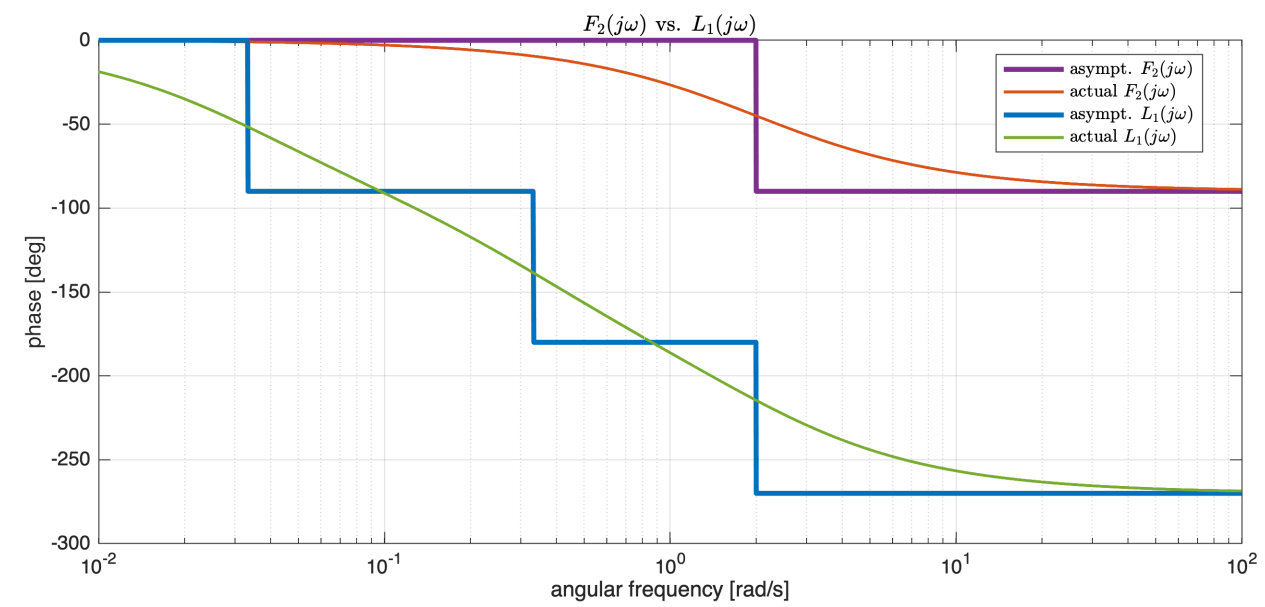
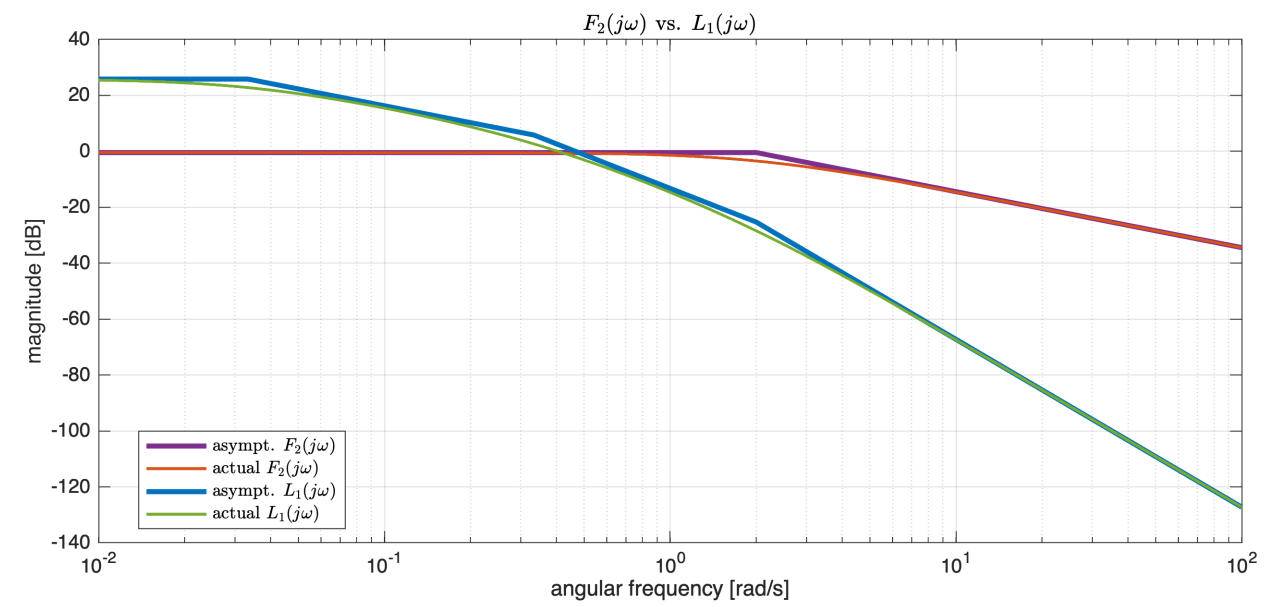
omVALS = logspace(-2, 2, 1e5); % ang. freqs in [1/100; 100]
[hax1, hax2] = drawBodediagrams(F2s, omVALS, Bcolors(3, :), 2.5, '-', ...
                                Bcolors(6, :), 1.5, '-', hf3); % assignment A1.2

drawBodediagrams(L1s_A1_4, omVALS, Bcolors(1, :), 2.5, '-', ...
                 Bcolors(4, :), 1.5, '-', hf3);

title(hax1, '$F_2(j \omega)$ vs. $L_1(j \omega)$', 'Interpreter', 'latex');
title(hax2, '$F_2(j \omega)$ vs. $L_1(j \omega)$', 'Interpreter', 'latex');

legend(hax1, 'asympt. $F_2(j \omega)$', 'actual $F_2(j \omega)$', ...
        'asympt. $L_1(j \omega)$', 'actual $L_1(j \omega)$', 'Location', 'best', ...
        'Interpreter', 'latex');
legend(hax2, 'asympt. $F_2(j \omega)$', 'actual $F_2(j \omega)$', ...
        'asympt. $L_1(j \omega)$', 'actual $L_1(j \omega)$', 'Location', 'best', ...
        'Interpreter', 'latex');

```



A1: Final Tuning using the controlSystemDesigner App

For detailed information about `sisoinit( )`

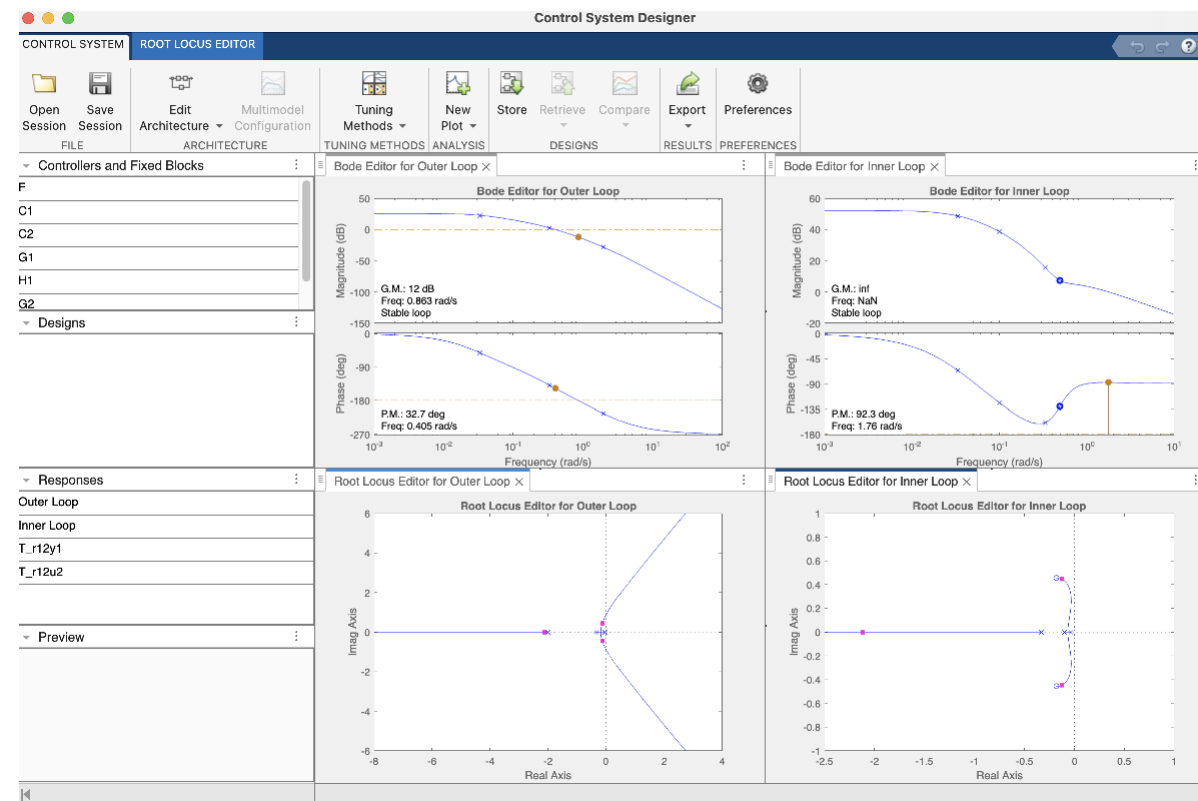
doc [sisoinit](#)

```

CSDconfig = sisoinit(6); %initialise the Control SYstem Designer with configuration 6
% filling the config structure with the TFs of the control system problem
% the plant TFs
CSDconfig.G1.Value = G1s;
CSDconfig.G2.Value = G2s;
% the controllers
CSDconfig.C1.Value = C1s_A1_4;
CSDconfig.C2.Value = C2s;
% the outer loop OL1
CSDconfig.OL1.Name = 'Outer Loop';
CSDconfig.OL1.View = {'bode', 'rlocus'};
% the inner loop OL2
CSDconfig.OL2.Name = 'Inner Loop';
CSDconfig.OL2.View = {'bode', 'rlocus'};
controlSystemDesigner(CSDconfig)

```

Every requirement is met. No further tuning is needed.



The Control System Designer session data are available in DESIGN_FA20240902_CSD_session_A1.mat

Second Approach: a Single-Loop Scheme

A2.1: The Open-Loop Transfer Function

In this scenario there is only a simple feedback loop, thus the open-loop transfer function becomes

$$\hat{L}(s) = C_3(s) \cdot G_1(s) \cdot G_2(s) \implies \hat{L}(s) = \frac{k_3}{(10s + 1) \cdot (30s + 1) \cdot (3s + 1)}$$

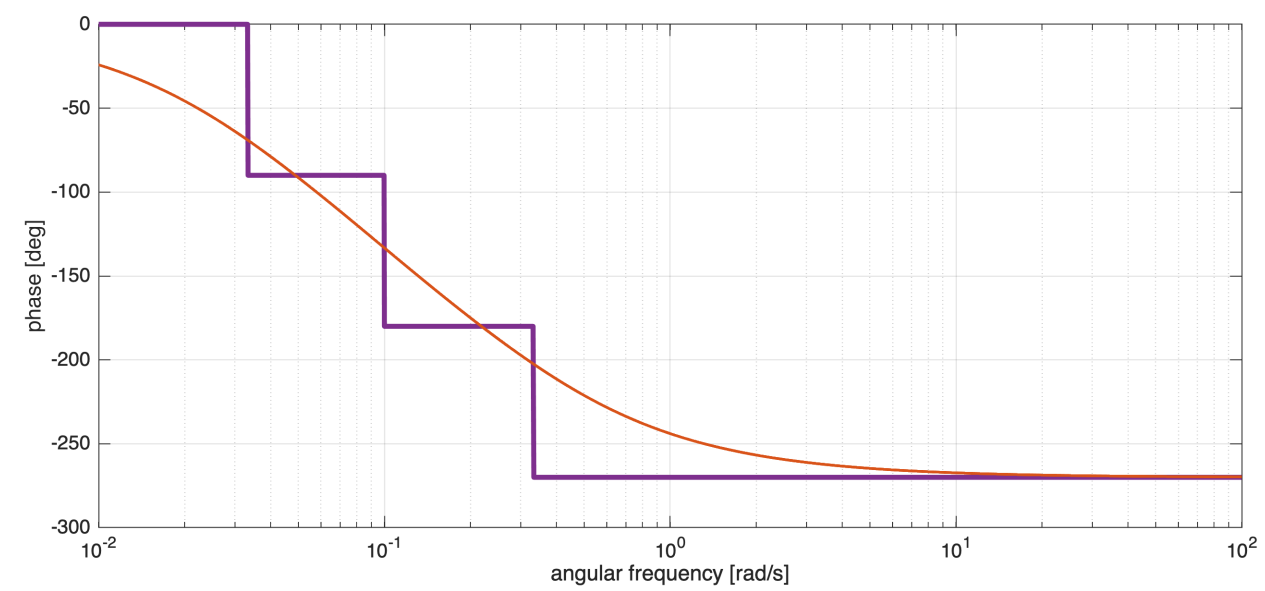
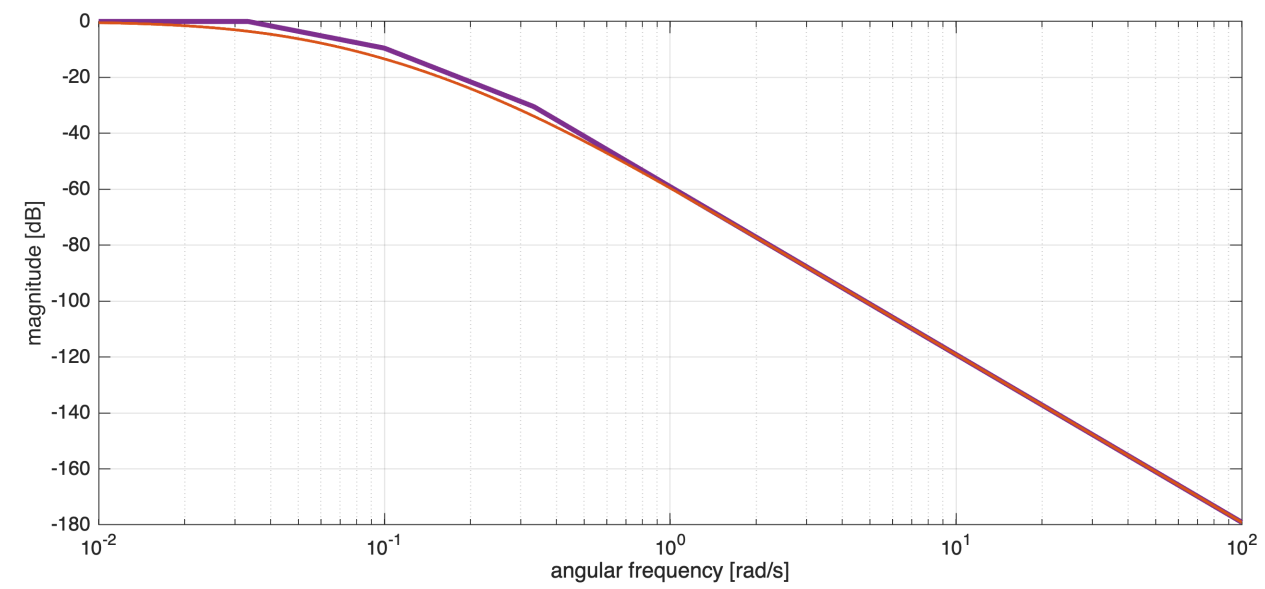
```
K3 = 1; % initial value for the controller gain
C3s = tf(K3); % the P controller

% --- the single-loop open-loop transfer function ---
hatLs = C3s * G1s * G2s;
```

A2.2: the Bode Plots of the Open-Loop Transfer Function

```
hf4 = figure('Units','centimeters', 'Position',[0.1,0.1, 26, 26]);

omVALS = logspace(-2, 2, 1e5); % ang. freqs in [1/100; 100]
[hax5, hax6] = drawBodediagrams(hatLs, omVALS, Bcolors(3, :), 2.5, '-', ...
                                Bcolors(6, :), 1.5, '-', hf4); % assignment A2.2
```

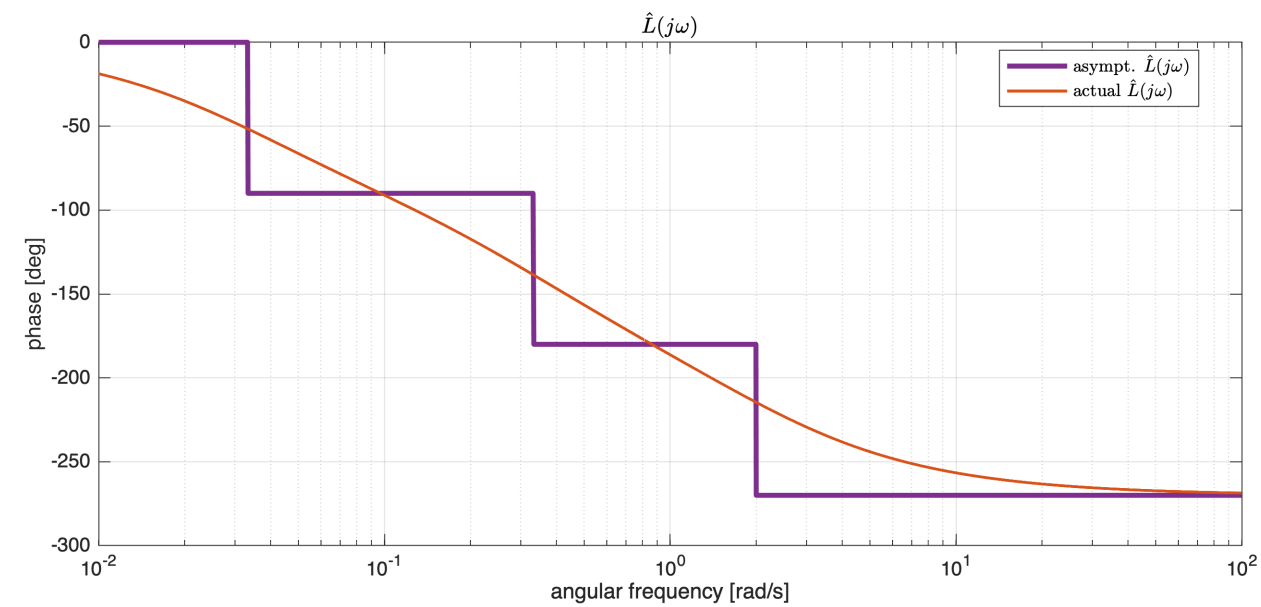
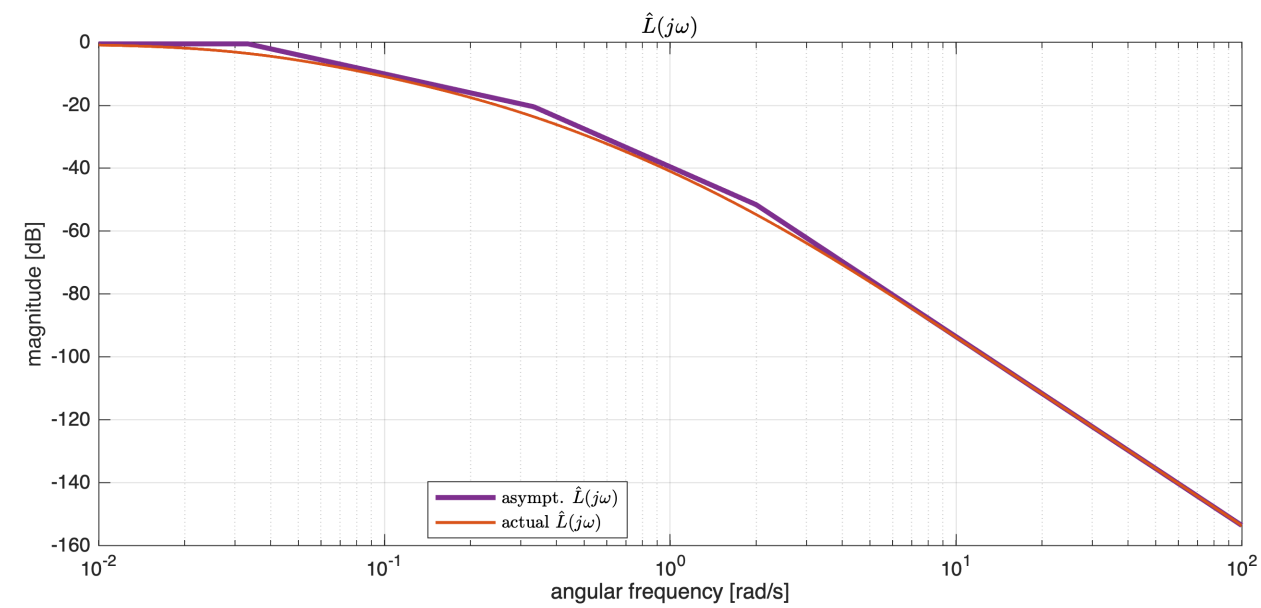


```

title(hax3, '$\hat{L}(j \omega)$', 'Interpreter', 'latex');
title(hax4, '$\hat{L}(j \omega)$', 'Interpreter', 'latex');

legend(hax3, 'asympt. $\hat{L}(j \omega)$', 'actual $\hat{L}(j \omega)$', ...
        'Location', 'best', ...
        'Interpreter', 'latex');
legend(hax4, 'asympt. $\hat{L}(j \omega)$', 'actual $\hat{L}(j \omega)$', ...
        'Location', 'best', ...
        'Interpreter', 'latex');

```

A2.3: The Comparison of the Open-Loop Frequency Responses in both the Scenarios

```
% ----- comparing L1s with hatLs -----
hf5 = figure('Units','centimeters','Position',[0.1,0.1, 26, 26]);

omVALS = logspace(-2, 2, 1e5); % ang. freqs in [1/100; 100]
[hax5, hax6] = drawBodeDiagrams(L1s, omVALS, Bcolors(3, :), 2.5, '-', ...
                                Bcolors(6, :), 1.5, '-', hf5); % assignment A1.2
```

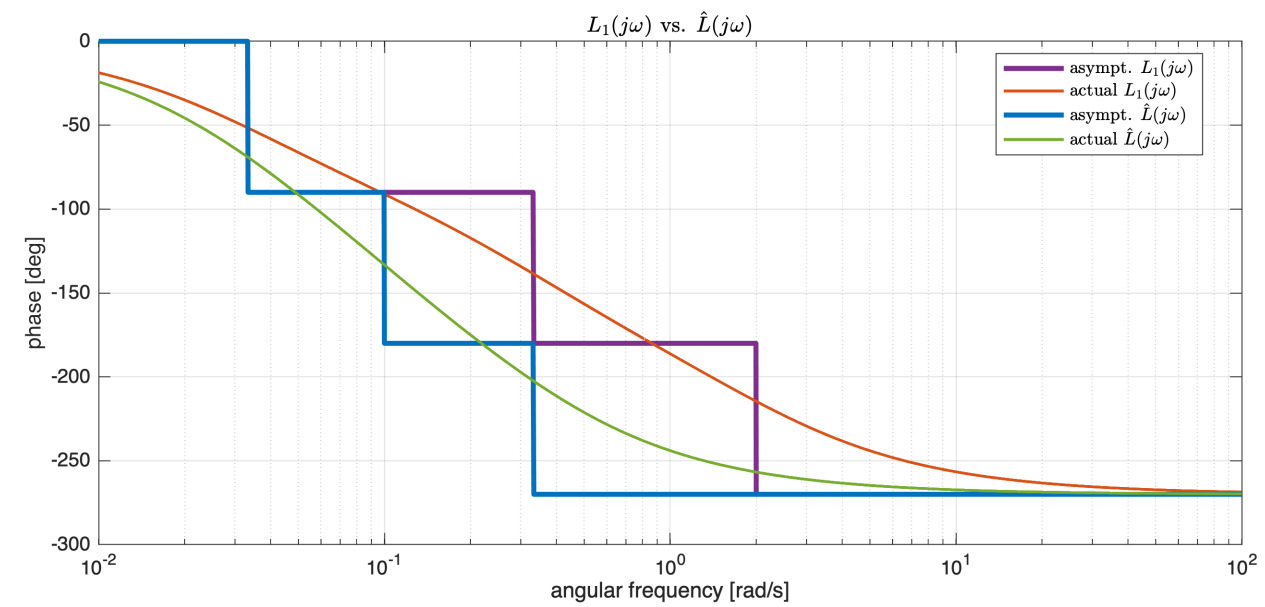
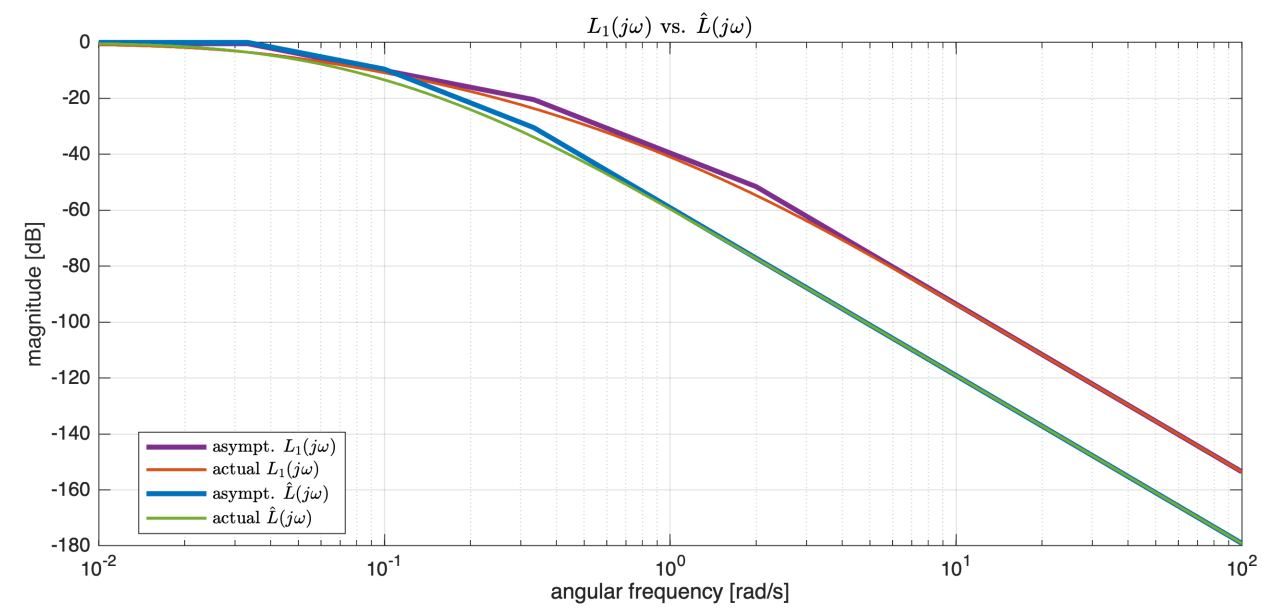
```

drawBodediagrams(hatLs, omVALS, Bcolors(1, :), 2.5, '-', ...
                Bcolors(4, :), 1.5, '-', hf5);

title(hax5, '$L_1(j \omega)$ vs. $\hat{L}(j \omega)$', 'Interpreter', 'latex');
title(hax6, '$L_1(j \omega)$ vs. $\hat{L}(j \omega)$', 'Interpreter', 'latex');

legend(hax5, 'asympt. $L_1(j \omega)$', 'actual $L_1(j \omega)$', ...
        'asympt. $\hat{L}(j \omega)$', 'actual $\hat{L}(j \omega)$', 'Location', 'best', ...
        'Interpreter', 'latex');
legend(hax6, 'asympt. $L_1(j \omega)$', 'actual $L_1(j \omega)$', ...
        'asympt. $\hat{L}(j \omega)$', 'actual $\hat{L}(j \omega)$', 'Location', 'best', ...
        'Interpreter', 'latex');

```



Remark: To point out some important different features, let's evaluate the stability margin for both the transfer functions $L_1(s)$ (with $k_1 = 1$ and $k_2 = 19$) and $\hat{L}(s)$ (with $k_3 = 1$)

```
[G1, P1, OMG1, OMP1] = margin(L1s)

G1 =
82.4061
P1 =
Inf
OMG1 =
0.8628
OMP1 =
NaN

[Gh, Ph, OMgh, OMPH] = margin(hatLs)

Gh =
19.0667
Ph =
-180
OMgh =
0.2186
OMPH =
0
```

The open-loop transfer function $\hat{L}(s)$ performs worse than $L_1(s)$ in the first scenario.

A2.4: Tuning the Controller

Given the controller $C_3(s) = k_3$, the open-loop transfer function results

$$\hat{L}(s) = C_3(s) \cdot G_1(s) \cdot G_2(s) \implies \hat{L}(s) = \frac{k_3}{(10s + 1) \cdot (30s + 1) \cdot (3s + 1)}$$

In order to properly tune the controller, let's perform the same algorithm as in the [previous scenario](#):

- find the angular frequency $\hat{\omega}_\pi$, such that $\arg \hat{L}(j\hat{\omega}_\pi) = -180^\circ$. This frequency exists for sure, as pointed out by the [Bode plots](#) in A2.2.
- evaluate the frequency response magnitude $|\hat{L}(j\hat{\omega}_\pi)|$ when $k_3 = 1$
- calculate the proper gain value $\bar{k}_3$ such that the requirement on the gain margin is met.

First approach: exploiting the Matlab command `margin( )`

```
[~, ~, omega_pi_approx, ~] = margin(hatLs)

omega_pi_approx =
0.2186

rad2deg(angle(freqresp(hatLs, omega_pi_approx)))

ans =
179.9999
```

Second approach: applying some mathematical manipulations.

As suggested, let's calculate the angular frequency $\hat{\omega}_\pi$ as

$$\hat{\omega}_\pi : \arg \hat{L}(j\hat{\omega}_\pi) = -180^\circ \iff \arg(k_3) - \arg[(10j\hat{\omega}_\pi + 1)(30j\hat{\omega}_\pi + 1)(3j\hat{\omega}_\pi + 1)] = -180^\circ$$

Assuming $k_3 > 0$, we obtain

$$\hat{\omega}_\pi : \arg \hat{L}(j\hat{\omega}_\pi) = -180^\circ \iff -\arg[(10j\hat{\omega}_\pi + 1)(30j\hat{\omega}_\pi + 1)(3j\hat{\omega}_\pi + 1)] = -180^\circ$$

Thus

$$\hat{\omega}_\pi : \arg \hat{L}(j\hat{\omega}_\pi) = -180^\circ \iff \arg[(10j\hat{\omega}_\pi + 1)] + \arg[(30j\hat{\omega}_\pi + 1)] + \arg[(3j\hat{\omega}_\pi + 1)] = 180^\circ$$

$$\arg[(30j\hat{\omega}_\pi + 1)] + \arg[(3j\hat{\omega}_\pi + 1)] = 180^\circ - \arg[(10j\hat{\omega}_\pi + 1)]$$

Since $\hat{\omega}_\pi > 0$

$$\arctan\left(\frac{30\hat{\omega}_\pi}{1}\right) + \arctan\left(\frac{3\hat{\omega}_\pi}{1}\right) = 180^\circ - \arctan\left(\frac{10\hat{\omega}_\pi}{1}\right) \iff \arctan(30\hat{\omega}_\pi) + \arctan(3\hat{\omega}_\pi) = 180^\circ - \arctan(10\hat{\omega}_\pi)$$

Given the trigonometric identities

$$\tan(\alpha + \beta) = \frac{\tan(\alpha) + \tan(\beta)}{1 - \tan(\alpha)\tan(\beta)} \quad \alpha + \beta \neq 90^\circ + n180^\circ, n \in \mathbb{Z}$$

and

$$\tan(180^\circ - \delta) = -\tan \delta$$

we obtain

$$\arctan(30\hat{\omega}_\pi) + \arctan(3\hat{\omega}_\pi) = 180^\circ - \arctan(10\hat{\omega}_\pi) \iff \frac{33\hat{\omega}_\pi}{1 - 90\hat{\omega}_\pi^2} = -10\hat{\omega}_\pi$$

Assume

$$\hat{\omega}_\pi \neq 0, \quad 1 - 90\hat{\omega}_\pi^2 \neq 0, \quad \hat{\omega}_\pi > 0$$

Thus

$$\frac{33\hat{\omega}_\pi}{1 - 90\hat{\omega}_\pi^2} = -10\hat{\omega}_\pi \iff \frac{33}{1 - 90\hat{\omega}_\pi^2} = -10 \iff \frac{33}{10} = -(1 - 90\hat{\omega}_\pi^2) \iff 3.3 = 90\hat{\omega}_\pi^2 - 1 \iff 90\hat{\omega}_\pi^2 = \frac{43}{10} \iff \hat{\omega}_\pi^2 = \frac{43}{900}$$

Finally

$$\omega_\pi = +\frac{\sqrt{43}}{30}$$

```
syms hat_omega
assume(hat_omega >=0)

OMeq_hat = hat_omega^2 == (43/900);
hat_omega_pi = solve(OMeq_hat, hat_omega)
```

```
hat_omega_pi =
sqrt(43)
30
```

```
hat_omega_pi_val = double(hat_omega_pi)
```

```
hat_omega_pi_val =
0.2186
```

```
rad2deg(angle(freqresp(hatLs,hat_omega_pi_val )))
```

```
ans =
180
```

Now, let's evaluate the **magnitude** of $\hat{L}(j\hat{\omega}_\pi)$:

Assuming $k_3 = 1$ we can calculate the magnitude of the frequency response $|\hat{L}(j\hat{\omega}_\pi)|$ as follows

```
mag_hatL_1 = abs(freqresp(hatLs,hat_omega_pi_val ))
```

```
mag_hatL_1 =
0.0524
```

To fulfil A2.4, we have to increase $|\hat{L}(j\hat{\omega}_\pi)|$:

```
gainM_target_dB = +12; % dB
gainM_target = 10^(gainM_target_dB/20)
```

```
gainM_target =
3.9811
```

In fact (refer to Part 9 of the course material, slide 24)

$$K_m = \frac{1}{|L(j\omega_\pi)|}$$

Thus, to obtain $\bar{K}_{m\text{ dB}} = +12\text{ dB}$, the magnitude $|\bar{L}_3(j\hat{\omega}_\pi)|$ should have the value

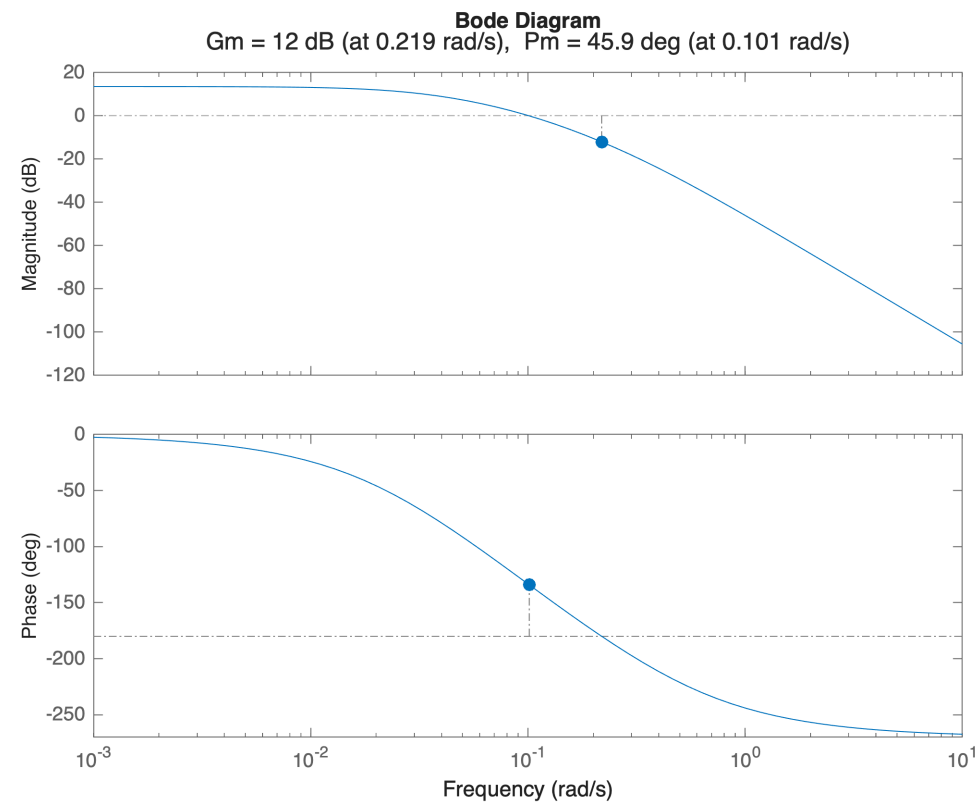
$$|\bar{L}_3(j\hat{\omega}_\pi)| = \frac{1}{\bar{K}_m} \approx \frac{1}{4} \implies \bar{k}_1 = \frac{|\bar{L}_3(j\hat{\omega}_\pi)|}{|L_3(j\hat{\omega}_\pi)|}$$

```
K3_A2_4 = (1/4)/mag_hatL_1
```

```
K3_A2_4 =
4.7667
```

Let's check the effective gain margin, when using this value for the gain k_1 :

```
C3s_A2_4 = tf(K3_A2_4);  
hatLs_A2_4 = C3s_A2_4*G1s*G2s;  
figure; margin(hatLs_A2_4)
```



A2.5: Crossover Frequency and Phase Margin

Simply

```
[gainM_1loop, phaseM_1loop, OM_GM_1loop, OM_PM_1loop]= margin(hatLs_A2_4)
```

```
gainM_1loop =  
4.0000  
phaseM_1loop =  
45.8954  
OM_GM_1loop =  
0.2186  
OM_PM_1loop =  
0.1014
```

Second Approach: a Recap

```
% --- the single loop controller C3s ---  
C3s_A2_4 = tf(K3_A2_4);  
  
% let's compute the open-loop TF of the inner loop L2s  
hatLs_A2_4 = C3s_A2_4*G1s*G2s;
```



```

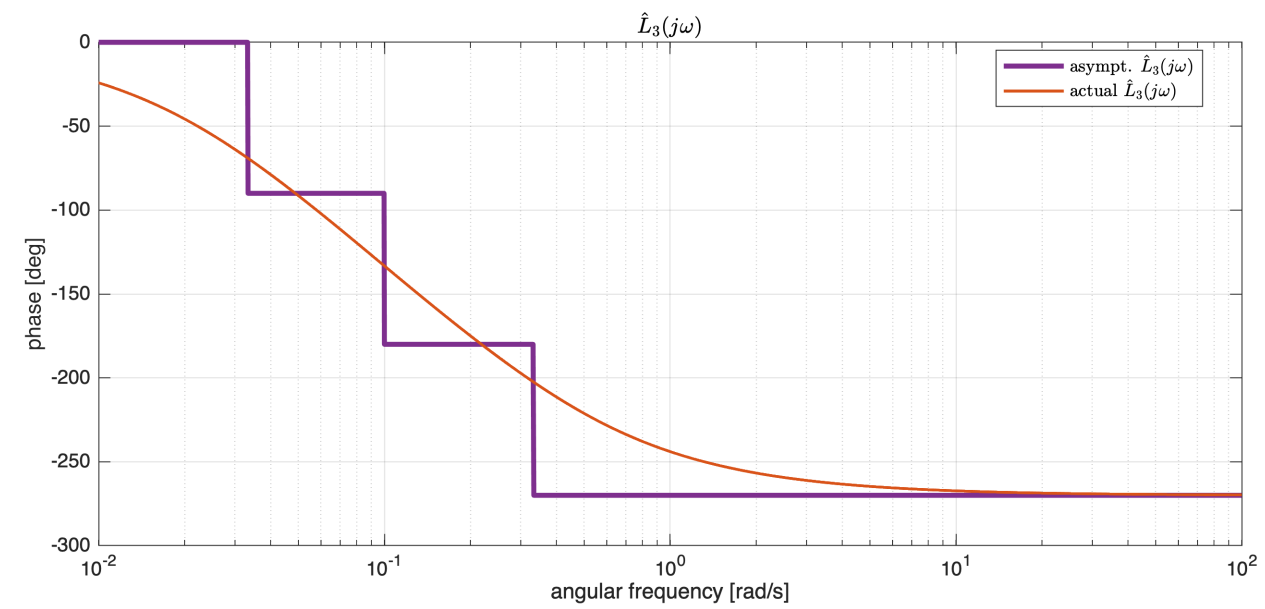
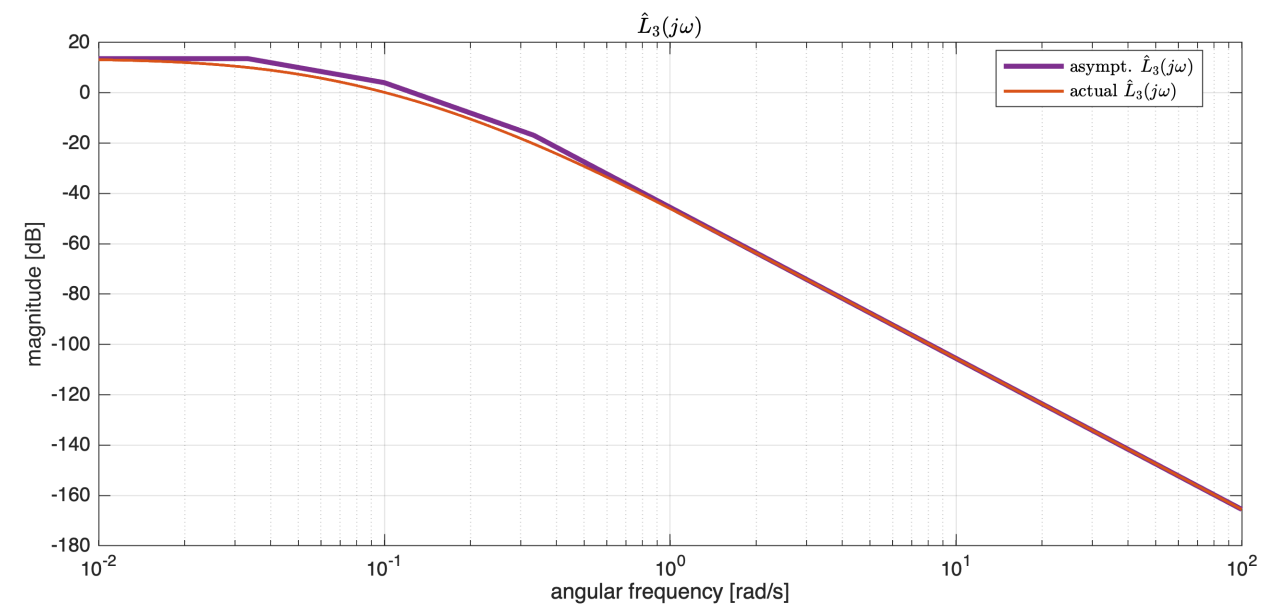
% ----- final Bode plots -----
hf8 = figure('Units','centimeters', 'Position',[0.1,0.1, 26, 26]);

omVALS = logspace(-2, 2, 1e5); % ang. freqs in [1/100; 100]
[hax1, hax2] = drawBodediagrams(hatLs_A2_4, omVALS, Bcolors(3, :), 2.5, '-', ...
                                Bcolors(6, :), 1.5, '-', hf8); % assignment A1.2

title(hax1, '$\hat{L}_3(j \omega)$', 'Interpreter', 'latex');
title(hax2, '$\hat{L}_3(j \omega)$', 'Interpreter', 'latex');

legend(hax1, 'asympt. $\hat{L}_3(j \omega)$', 'actual $\hat{L}_3(j \omega)$', ...
        'Location', 'best', ...
        'Interpreter', 'latex');
legend(hax2, 'asympt. $\hat{L}_3(j \omega)$', 'actual $\hat{L}_3(j \omega)$', ...
        'Location', 'best', ...
        'Interpreter', 'latex');

```



A2: Final Tuning using the controlSystemDesigner App

For detailed information about `sisoinit( )`

doc `sisoinit`

```
CSDconfig = sisoinit(1) %initialise the Control SYstem Designer with configuration 1
```

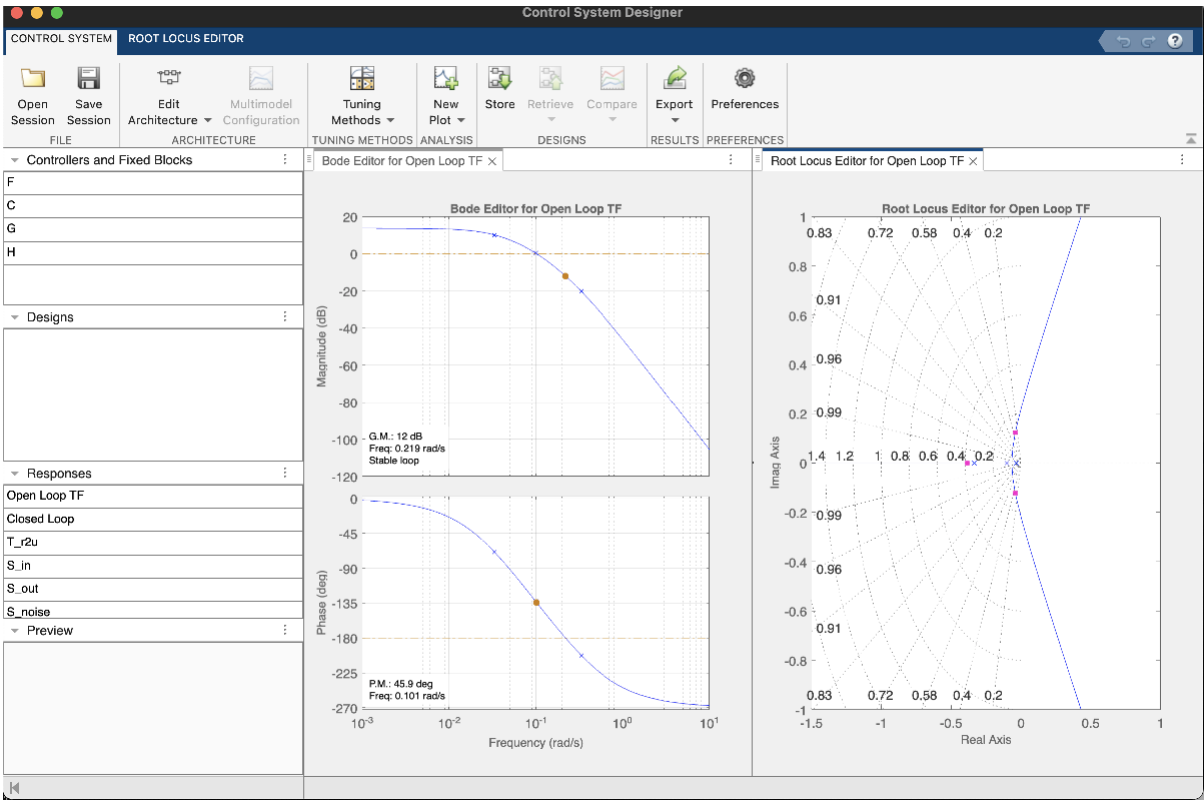


```
Name: ''
Description: 'Design snapshot.'
FeedbackSign: -1
Input: {4x1 cell}
Output: {2x1 cell}
Version: 2
Configuration: 1
LoopView: [10x1 sisodata.looptransfer]
C: [1x1 sisodata.TunedZPKSnapshot]
H: [1x1 sisodata.system]
G: [1x1 sisodata.system]
OL1: [1x1 sisodata.TunedLoopSnapshot]
F: [1x1 sisodata.TunedZPKSnapshot]
CL1: [1x1 sisodata.TunedLoopSnapshot]
```

```
% filling the config structure with the TFs of the control system problem
% the plant TF
CSDconfig.G.Value = G1s*G2s;
% the controller
CSDconfig.C.Value = C3s_A2_4;
% the open loop OL1
CSDconfig.OL1.Name = 'Open Loop TF';
CSDconfig.OL1.View = {'bode','rlocus'};
```

```
controlSystemDesigner(CSDconfig)
```

Every requirement is met. No further tuning is needed.



The Control System Designer session data are available in DESIGN_FA20240902_cSD_session_A2.mat

A2.6: Final Comparison

```
% ----- comparing the scenarios -----
```

```

hf10 = figure('Units','centimeters', 'Position',[0.1,0.1, 26, 26]);

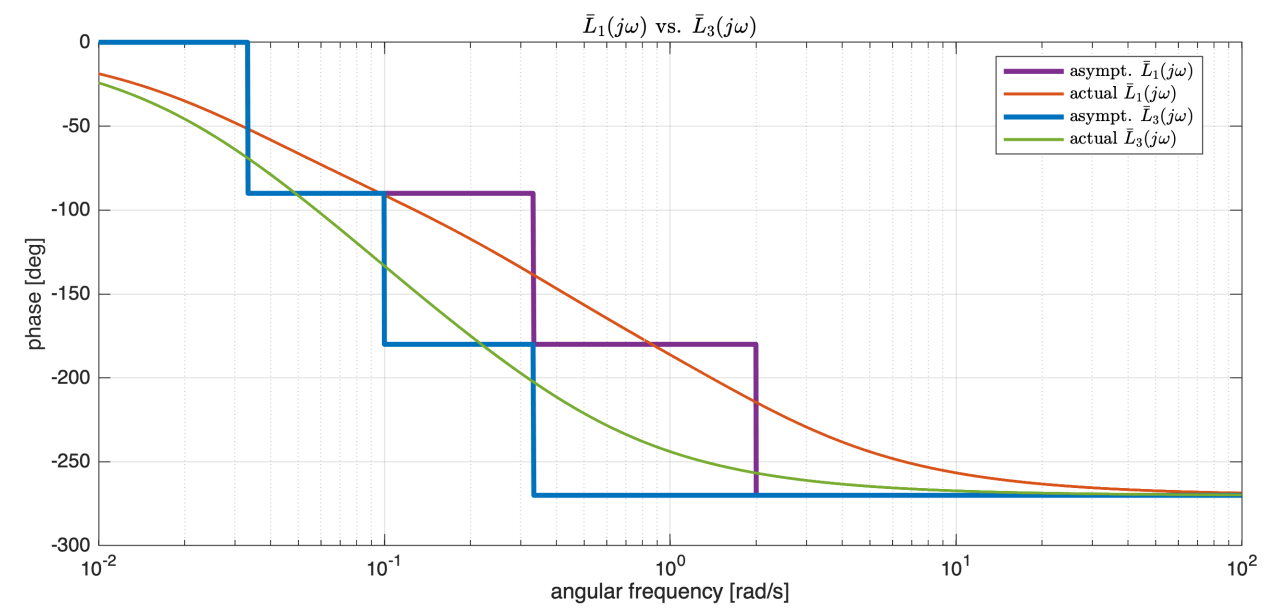
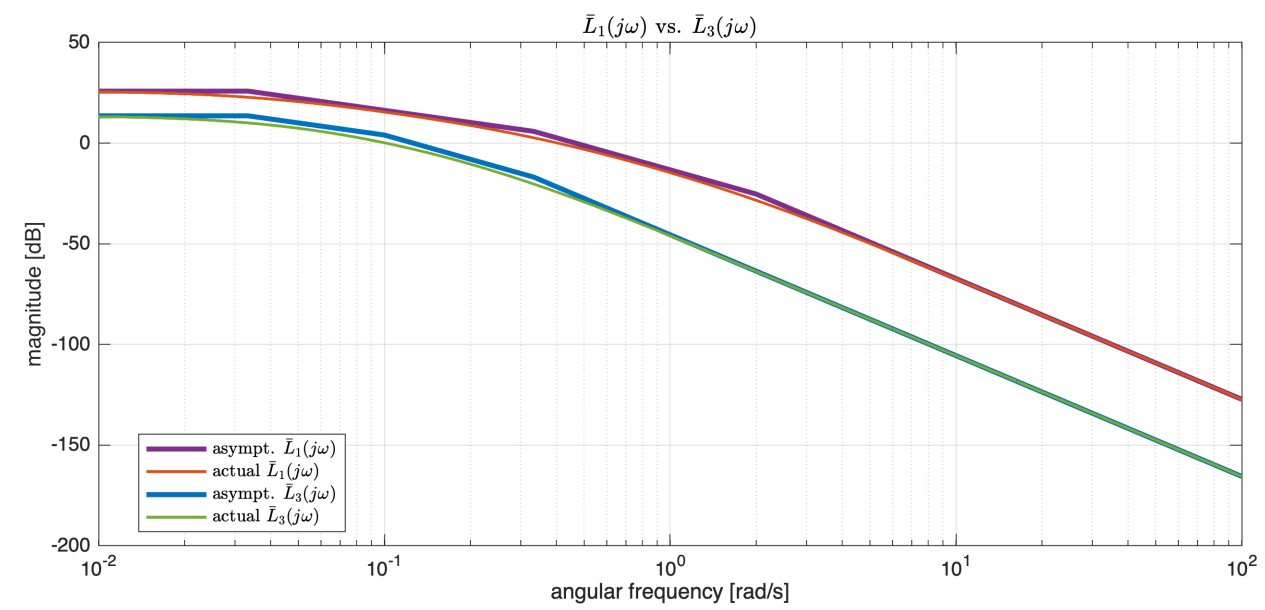
omVALS = logspace(-2, 2, 1e5); % ang. freqs in [1/100; 100]
[hax5, hax6] = drawBodediagrams(L1s_A1_4, omVALS, Bcolors(3, :), 2.5, '-', ...
    Bcolors(6, :), 1.5, '-', hf10); % assignment A1.2

drawBodediagrams(hatLs_A2_4, omVALS, Bcolors(1, :), 2.5, '-', ...
    Bcolors(4, :), 1.5, '-', hf10);

title(hax5, '$\bar{L}_1(j \omega)$ vs. $\bar{L}_3(j \omega)$', 'Interpreter', 'latex');
title(hax6, '$\bar{L}_1(j \omega)$ vs. $\bar{L}_3(j \omega)$', 'Interpreter', 'latex');

legend(hax5, 'asympt. $\bar{L}_1(j \omega)$', 'actual $\bar{L}_1(j \omega)$', ...
    'asympt. $\bar{L}_3(j \omega)$', 'actual $\bar{L}_3(j \omega)$', 'Location', 'best', ...
    'Interpreter', 'latex');
legend(hax6, 'asympt. $\bar{L}_1(j \omega)$', 'actual $\bar{L}_1(j \omega)$', ...
    'asympt. $\bar{L}_3(j \omega)$', 'actual $\bar{L}_3(j \omega)$', 'Location', 'best', ...
    'Interpreter', 'latex');

```



% 1st scenario

```
[gainM_2loop, phaseM_2loop, OM_GM_2loop, OM_PM_2loop]= margin(L1s_A1_4)
```

```
gainM_2loop =
4.0001
phaseM_2loop =
32.7168
OM_GM_2loop =
0.8628
OM_PM_2loop =
0.4050
```

% 2nd scenario

```
[gainM_1loop, phaseM_1loop, OM_GM_1loop, OM_PM_1loop]= margin(hatLs_A2_4)
```

```
gainM_1loop =  
4.0000  
phaseM_1loop =  
45.8954  
OM_GM_1loop =  
0.2186  
OM_PM_1loop =  
0.1014
```

```
% DC gain – 1st scenario  
dcgain(L1s_A1_4)
```

```
ans =  
19.5708
```

```
% DC gain – 2nd scenario  
dcgain(hatLs_A2_4)
```

```
ans =  
4.7667
```

```
% closed-loop step response – scenario 1  
F1s = feedback(L1s_A1_4, 1, -1);  
stepinfo(F1s, "SettlingTimeThreshold", 0.01)
```

```
ans = struct with fields:  
    RiseTime: 2.8796  
    TransientTime: 36.9733  
    SettlingTime: 36.9733  
    SettlingMin: 0.7909  
    SettlingMax: 1.3368  
    Overshoot: 40.5129  
    Undershoot: 0  
        Peak: 1.3368  
    PeakTime: 7.3786
```

```
% closed-loop step response – scenario 1  
hatFs = feedback(hatLs_A2_4, 1, -1);  
stepinfo(hatFs, "SettlingTimeThreshold", 0.01)
```

```
ans = struct with fields:  
    RiseTime: 11.2110  
    TransientTime: 112.5913  
    SettlingTime: 112.5913  
    SettlingMin: 0.7288  
    SettlingMax: 1.1023  
    Overshoot: 33.3557  
    Undershoot: 0  
        Peak: 1.1023  
    PeakTime: 28.4032
```

Final remarks: The **second approach** (a single feedback loop) solution **performs worse** than the solution in the former scenario (dual-loop control scheme):

- The gain margins are equal by design.
- In the latter case, the static gain is lower than in the former scenario: the steady-state error to a stepwise input will be larger in the second scenario than in the first one.
- Also, the crossover frequency is lower in the latter case: the closed-loop step response settling time is larger than the one in the former case (approximately three times the settling time obtained in the first scenario).

- As a trade-off to the faster closed-loop step response, the overshoot in the step response in the first scenario is higher than the overshoot in the latter scenario (approximately 7 % higher).

Removing the added folders to the Matlab path

```
% remove folders (and subfolders) to search path  
rmpath(genpath('BodeDiagram/'))
```